Machine Learning Foundations - Supervised & Unsupervised Mo

Course: Machine Learning | Semester 7 | Contributor: Alex Chen

Gradient descent, logistic regression, SVMs, decision trees, random forests, and k-means cluste

Supervised Learning Algorithms

* Linear regression with mean squared error MSE cost function and gradient descent optimization.
* Logistic regression, sigmoid activation, and cross-entropy loss function.
* Support Vector Machines SVM: maximum margin hyperplane, soft margin slack, kernel trick.

Model Evaluation & Overfitting

* Bias-Variance tradeoff: high bias underfitting vs high variance overfitting.
* Regularization techniques: L1 Lasso, feature selection and L2 Ridge, weight shrinkage.
* Evaluation metrics: Confusion matrix, Precision, Recall, F1-Score, and ROC-AUC curve.

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